

SSVV 2024-2025 Take-Home-B Exam Subject

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Task summary:

Task 1: Test Case Design and Bug Fix

We analyzed Version A of the provided Java code (PaymentProcessor.java). We designed a JUnit test suite covering all combinations of: payment methods (CREDIT_CARD, PAYPAL, CASH) and first-order status (true, false). The test design technique used is black-box testing: to validate expected outputs against different inputs. We identified a bug: the tax (15%) was applied before discounts. We fixed the bug by correctly applying the tax after the discount. Created Version B with the corrected logic.

Task 2: Execution on Both Versions

We executed the same test suite on Version A and Version B. Tests revealed incorrect final amounts in Version A (due to tax order). Version B passed all tests and maintained expected delivery fee behavior. We validated that the bug fix did not break existing functionality.

Task 3: Regression Test Selection

We selected only tests related to processPayment() for re-execution. Tests for calculateDeliveryFee() and exception handling were skipped, as the method logic was unchanged. Our regression test selection was based on: impact-based analysis(only retested the code paths affected by the change). We ensured that each discount scenario was retested post-fix.